PulmoTwin | System Architecture

Multimodal cardiopulmonary digital twin for proactive adverse-event forecasting and personalised clinical intervention

Digital Twin Challenge 2026 - Happiest Health Summit
Chronic asthma / cardiopulmonary care, India
Data flows left to right

1 DATA SOURCES
Multimodal inputs

ABDM / HL7 FHIR R4 EHR
digital_twin/ehr_connector.py

- Longitudinal static baseline record
- Spirometry (FEV1, personal-best PEF), FeNO, IgE, blood eosinophils
- Diagnosis (ICD-10), comorbidities, regimen, past exacerbations
- Synthetic data modelled on Synthea / MIMIC-IV, zero PII

Wearable IoT telemetry
digital_twin/wearable_sim.py

- Continuous PPG: heart rate, HRV rMSSD, SpO2, respiratory rate
- Peak flow (PEF) and FEV1 readings
- Smart inhaler actuation over BLE (puff count and timing)
- Simulated sensor stream in the prototype

Atmospheric telemetry
Open-Meteo air quality + weather APIs

- Live GPS sync with 48-hour hourly forecast
- US AQI, PM2.5, PM10, NO2, SO2, O3, CO2
- Temperature and humidity
- City sensor grid feeds the physician view

Patient input
web_ui/index.html

- Bilingual English / Hindi intake, voice and text
- Symptom and PEF diary: wheeze, cough, chest tightness, reliever puffs
- 3-step assessment wizard, manual entry fallback
- Free text is safety-scanned before any model sees it

2 INGESTION AND BACKEND
Fusion, API, storage

Connectors and adapters
ehr_connector.py, wearable_sim.py

- FHIR R4 bundle mapped to the internal patient schema
- Wearable and inhaler events normalised into time series
- Open-Meteo data fetched per GPS location
- data_generator.py builds the synthetic cohort

Preprocessing and fusion
preprocess.py, utils.py

- Cleans and encodes clinical, sensor and air-quality features
- Fuses the three streams into one multimodal feature vector
- Fitted pipeline saved in preprocessing.pkl

Flask REST API
app.py

- Flask 3.1 served by Gunicorn
- /api/digital-twin/* endpoints, including simulate
- Token authentication and IDOR guards
- Serves the patient portal and physician console

Persistence
SQLite via Flask-SQLAlchemy

- Profiles, sensor readings, diary entries, agent plans
- Tamper-evident audit trail of every action
- Stored locally, never sold or shared without consent

3 DIGITAL TWIN ENGINE
Simulate, predict, test

Physiological twin
digital_twin/twin_engine.py

- Airway resistance: Raw = 1.8 / PEF_ratio^1.6
- Lung compliance: Cdyn = 70 x PEF_ratio - FeNO/10
- Alveolar-arterial O2 (A-a DO2) gradient
- Autonomic sympathetic tone from HRV rMSSD
- Exacerbation Risk Index (ERI) from 0 to 1

Multimodal risk ensemble
models.py

- Soft-voting stack: Logistic Regression (w1), Random Forest, 100 trees (w2), Gradient Boosting, 150 stages (w2)
- Accuracy 73.0%, Brier 0.1393 on N=300 held-out
- Shannon-entropy uncertainty rejects low-confidence output

24-hour forward projection
twin_engine.py (ODE kinetics)

- Differential kinetics of airway inflammation
- Forecasts Raw surges and exacerbations 8 to 24 h before onset
- ERI tiers: green controlled, yellow caution, red critical
- Feeds the 48-h safe-window timeline

Counterfactual what-if sandbox
digital_twin/counterfactual.py

- Tests interventions in silico before acting
- Indoor HEPA filter: PM2.5 down 75%, risk down 45.2%
- Proactive SMART inhaler: risk down 66.7%, ERI 0.84 to 0.28
- Shifts evening run to the 06:30-08:30 window

4 SAFETY RAILS
Deterministic, no black box

Acute distress interceptor
input guard, runs first

- 61 distress phrases and 32 physical signs
- English, Hindi and Hinglish, negation-aware
- Bypasses the model: instant SOS, dial 108 / 112
- Also fires when PEF drops below 50% of best

GINA clinical override
GINA Step-5 rules

- Daily symptoms or frequent night waking force High Risk
- High-risk recall rises 72.09% to 82.95% (107 of 129)
- Deliberate bias: a false alarm costs less than a miss

Output sanitizer
post-generation filter

- Regex check on every agent reply
- Blocks dosage changes and diagnostic claims
- Swaps violations for a clinical fallback disclaimer

Zero-trust approval queue
human in the loop

- Agent never sends SMS or edits external systems itself
- Every action stays PENDING until a human approves
- Clinician sign-off queue for physician-side plans

5 INTERFACES AND OUTPUTS
Patient and clinician

Patient portal
web_ui/index.html

- Twin cockpit: live telemetry, organ biomechanics, What-If
- Autonomous agent: 48-h air-quality timeline, voice and text assistant, symptom diary
- Patient assessment: risk score with action plan

Physician console
view-admin

- Cohort triage table with per-model scores
- KPI cards: cohort, sensor nodes, inferences, alerts
- Clinician sign-off queue and Quick Prescribe (GINA)
- Broadcast advisory to the cohort

Reports and privacy
ReportLab 5.0

- 30-day GINA consultation summary PDF
- JSON export for data portability
- Permanent account and health-record erasure
- Aligned with DPDP Act 2023 and GDPR principles

Deployment targets
Pages, Render, Vercel, Docker

- GitHub Pages: in-browser physiology engine
- Render, Vercel, Docker and Procfile for the Flask app
- pytest suite covers guardrails, agent and API

TECHNOLOGY STACK
Icons are simplified vector marks, not official brand logos

- **Python 3.11** — core language
- **Flask 3.1** — REST API, Gunicorn
- **SQLite** — SQLAlchemy ORM
- **NumPy, SciPy** — ODE twin engine
- **scikit-learn** — stacking ensemble
- **ReportLab** — PDF reports
- **HL7 FHIR R4** — ABDM schemas
- **Open-Meteo** — air quality, weather
- **Docker** — container, Render
- **HTML5, ES6 JS** — web cockpit

PulmoTwin is an investigational decision-support research prototype, not a certified medical device. In an emergency dial 112 / 108.
Source: github.com/Kabirroy12345/Digital_Twin | MIT license | Metrics from README held-out evaluation (N=300)